

PHYS 427: Lecture # 12

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Lecture Span

1. Grand Canonical Ensemble

For midterm, Ω, S, T, Z, F . Up to and including lecture 10.

Grand Canonical Ensemble

Reservoir and system are coupled. Note that U_{tot} and N_{tot} are fixed. Also, S is in a single microstate i with energy ϵ_i , N_i particles.

The probability for S to be in that state is

$$p(N_i, \epsilon_i) = \frac{\Omega_{R+S}(N_i, \epsilon_i)}{\sum \Omega_{R+S}} \quad (1)$$

Moreover,

$$\Omega_{R+S} = \Omega_R \Omega_S \quad (2)$$

Then,

$$\frac{p(N_1, \epsilon_1)}{p(N_2, \epsilon_2)} = \frac{\Omega_R(N_0 - N_1, U_0 - \epsilon_1)}{\Omega_R(N_0 - N_2, U_0 - \epsilon_2)} \quad (3)$$

$$= \frac{\exp(S_R(\dots)/k_B)}{\exp(S_R(\dots)/k_B)} \quad (4)$$

$$= \exp((S_R(N_1) - S_R(N_2))/k_B) \quad (5)$$

Then, expand about U_0 and N_0 , then

$$S_R \approx S_R(U_0, N_0) - \frac{\partial S_R}{\partial N} \cdot N_1 - \frac{\partial S}{\partial U} \cdot \epsilon_1 \quad (6)$$

$$= S_R + \frac{\mu}{T} N_1 - \frac{\epsilon_1}{T} \quad (7)$$

Then,

$$\frac{p_1}{p_2} = \frac{\exp\left(\frac{\mu N_1 - \epsilon_1}{k_B T}\right)}{\exp\left(\frac{\mu N_2 - \epsilon_2}{k_B T}\right)} \quad (8)$$

Moreover, we know that

$$\sum_i \frac{p_i}{p_2} = \frac{1}{p_2} \quad (9)$$

Therefore,

$$p_2 = \frac{\exp(-(\epsilon_1 - \mu N_1)/k_B T)}{\mathcal{Z}} \quad (10)$$

Here, $\mathcal{Z}$ is the Gibbs sum and defined as:

$$\mathcal{Z} = \sum_{N_i, \epsilon_i(N_i)} e^{-(\epsilon_i - \mu N_i)/k_B T} \quad (11)$$

The top is called the Gibbs factor. Also,

$$\mathcal{Z} = \sum_N \exp(\mu N/k_B T) Z(N, T) \quad (12)$$

Also,

$$\langle N \rangle = \lambda \frac{\partial}{\partial \lambda} \ln(\mathcal{Z}(\lambda, T)) \quad (13)$$

Where

$$\lambda = \exp(\mu/k_B T) \quad (14)$$

$$\langle \epsilon \rangle = \frac{\mu}{\beta} \partial_\mu \ln \mathcal{Z} - \partial_\beta \ln \mathcal{Z} \quad (15)$$

Next, we calculate

1. $\mathcal{Z}$ for myoglobin
2. probability that a single O_2 binds to it

Note, the # of O_2 binded to the myoglobin is either $N = 0$ or $N = 1$. Then, the gibbs sum is

$$\mathcal{Z} = 1 + \exp(-\beta(\epsilon_{O_2} - \mu)) \quad (16)$$

The prob. of 1 O_2 binding to the myoglobin is

$$f = \frac{1}{1 + \lambda^{-1} \exp(\epsilon_{O_2}/k_B T)} \quad (17)$$

Then, assume that O_2 is in equilibrium with the Mb and O_2 , then treat oxygen as an ideal gas and calculate

$$\mu(O_2) = k_B T \ln(n_{O_2}/n_Q) \quad (18)$$

Then,

$$f = \frac{p}{p + n_Q k_B T \exp(\epsilon_{n_Q}/k_B T)} \quad (19)$$

Here, p is the partial pressure of Oxygen.

Formalism

Assume two chemical species N_A^0 and N_B^0 , then

$$\mathcal{Z} = \sum_{N_A, N_B, \dots} \exp((\mu_A N_A + \dots)/k_B T) \sum_{\epsilon(N_A, N_B, \dots)} \exp(-\epsilon(N_A, N_B, \dots)/k_B T) \quad (20)$$

Carbon monoxide poisoning

Assume O_2 and CO in the air. They are competing to attach to your myoglobin. But CO_2 has a lower binding energy. The probability of CO_2 binding to Mb is greater than that of O_2 .